

# What the Model Stops Listening To: Explanation Displacement Under Perceptual Audio Coding in Speech Emotion Recognition

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**Abstract**—Speech emotion recognition is validated on uncompressed audio and deployed on coded audio. The gap is assumed benign because every ingestion path decodes to PCM before inference. We test that assumption on the quantity accuracy cannot see: the evidence a model actually uses. Two arms share one canonical reference signal derived from CREMA-D and one coding manipulation (MP3 at  $64\text{ kbit s}^{-1}$ ). Arm A probes a commercial multimodal audio language model through occlusion attribution over spectro-temporal regions; Arm B trains fifteen interpretable classifiers and networks on 436 named acoustic features and reads their evidence with SHAP, LIME, permutation importance, partial dependence, global surrogates, and six gradient-based attribution methods. Both arms are calibrated against a null: a 1LSB-dither perturbation in Arm A, a decode-path control in Arm B. Four results survive that calibration. First, coding relocates evidence more than it relocates accuracy: the language model shows no detectable loss of discriminability (AUC 0.734 to 0.746, all intervals overlapping) while its attribution rank concordance falls to  $\rho = 0.491$  against a measured null floor of 0.699 ( $p = 4.2 \times 10^{-8}$ ); in the feature arm, XGBoost loses 2.2 accuracy points on AAC yet replaces three of its twenty-five most-attributed features. Second, explanation stability is a property of the model family, not of the codec, and it is ordered by how a model consumes its inputs — tree ensembles  $\rho \geq 0.98$ , linear models 0.94–0.99, the neural network 0.677, the language model 0.491. Third, both arms independently localise the damage to 4 kHz to 8 kHz, and the same physical distortion produces opposite behavioural outcomes depending on whether a model thresholds that band or multiplies it. Fourth, the container is inert: substituting an MP3 for its bit-identical decoded WAV moves the language model’s posterior less than one least significant bit of noise ( $0.83\times$  the null floor), and re-wrapping AAC as WAV moves the fifteen feature models by a median of 0.003 balanced accuracy. Because Arm B’s evidence is named, the mechanism is diagnosable and repairable: masking a single feature, `spectral_contrast_6_mean`, recovers 98% of an RBF-SVM’s 38.3-point loss at no cost on clean audio. We argue that attribution stability is a reportable robustness property distinct from accuracy, that it requires a null perturbation to be interpretable at all, and that named-feature pipelines retain a diagnostic handle that end-to-end systems do not.

**Index Terms**—Speech emotion recognition, explainable artificial intelligence, perceptual audio coding, attribution stability, covariate shift, audio language models, model robustness, affective computing.

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Code, stimulus manifests, the feature table, and every result file underlying the tables and figures are released with this paper; see the Data Availability Statement.

## I. Introduction

AUDIO almost never reaches an inference endpoint in the representation in which it was captured. Contact-centre archives hold MP3; streaming clients negotiate Opus or AAC; broadcast pipelines resample and requantise; a clip may be transcoded three times between microphone and model. Evaluation practice runs the other way: benchmarks are distributed as uncompressed WAV and read directly from disk.

The gap is usually dismissed on a reasonable-sounding argument. Any ingestion path decodes to linear PCM before feature extraction, so the delivery format is an implementation detail with no behavioural consequence. Where the argument has been tested at all, it has been tested on accuracy, and accuracy is often reassuring: Reddy and Vijayarajan [1] report that codec effects on speech emotion recognition (SER) are real but modest and front-end-dependent.

Accuracy is the wrong instrument for the question we care about. A deployed affective system is increasingly required to say not only what it concluded but why — an attribution over acoustic evidence, surfaced to a clinician, an auditor, or a quality-assurance analyst. That second output has its own robustness properties, and they are not implied by the first. Ghorbani et al. [2] established for image classifiers that perceptually indistinguishable inputs receiving identical predictions can be assigned materially different attributions; Alvarez-Melis and Jaakkola [3] showed the same fragility more generally. Both used adversarially constructed perturbations. A transcode is not adversarial. It is routine, ubiquitous, and nobody logs it.

### A. Two estimands

The question therefore decomposes into two that admit different answers:

- 1) Does perceptual coding degrade task performance?
- 2) Does coding alter the evidential basis of the model’s output, independently of whether the output changes?

Answering the second requires a model whose evidence can be inspected, and inspectability is not a single property. It is a spectrum. At one end sits a commercial audio language model reachable only through a chat-completions endpoint, where attribution must be estimated by perturbing

the input. At the other sits a decision tree over named acoustic descriptors, where the evidence is the model. Prior work on codec robustness has occupied one end or the other. Neither end alone can separate what the codec does to the signal from what the model does with the codec.

## B. Contributions

This paper places both ends of that spectrum on a single controlled manipulation and reports what each can and cannot see.

- 1) A two-arm design on one canonical reference. Both arms derive every stimulus from the same EBU R128-normalised 16 kHz monophonic reference and share the MP3-at-64 kbit s<sup>-1</sup> contrast, so cross-arm statements are about model class rather than about differing signal chains (Section III).
- 2) Null calibration on both sides. Arm A calibrates its perturbation estimator against an inaudible 1 LSB dither; Arm B calibrates against a decode-path control in which container routing changes but the codec output does not. Every effect is reported relative to its own floor (Sections IV, V).
- 3) Evidence that explanation displacement exceeds accuracy displacement, and how far the claim extends. The strong dissociation — unchanged accuracy, changed evidence — is observed in Arm A but does not reproduce in Arm B. The weaker and still consequential form holds in both: the accuracy delta systematically understates the attribution delta (Section VI).
- 4) A mechanism that only the two arms together supply. Both independently localise coding damage to 4 kHz to 8 kHz. In Arm A that band carries modest reliance, so sensitivity survives; in Arm B the same band enters models that multiply it, and they collapse. One physical cause, two opposite behavioural outcomes, mediated by model family (Section VI).
- 5) A diagnosable and cheap repair, available only where features are named. Neutralising a single top-band descriptor recovers 98 % of an RBF-SVM’s loss and 92 % of a neural network’s, at no cost on uncompressed audio (Section V-F).
- 6) Two negative results retained rather than smoothed. Post-hoc attribution methods disagree substantially with each other on the same fitted model, and a global attribution ranking can report “nothing changed” while the classifier has stopped working (Sections V-E2, V-E4).

## II. Related Work

### A. Perceptual coding and speech task degradation

Degradation of speech systems under lossy coding is documented but shallowly characterised. Reddy and Vijayarajan [1] find that standard codecs reduce SER accuracy with a magnitude conditional on both codec

family and feature representation, which already implies that coding interacts with the front end rather than degrading performance uniformly.

Of greater methodological consequence is a result from neural codec benchmarking. Wu et al. [4] demonstrate that signal-domain fidelity metrics fail to predict the retention of task-relevant information: a codec can score well under reconstruction error while discarding paralinguistic content that downstream tasks require. We take this seriously enough to report signal fidelity and model behaviour as separate quantities throughout, treating neither as a surrogate for the other.

### B. Interpretability for tabular and audio models

Post-hoc attribution has converged on a small set of workhorses: local surrogate fitting [5], Shapley-value attribution [6] with an exact polynomial-time algorithm for tree ensembles [7], and occlusion sensitivity [8] for models exposed only through their input–output interface. For differentiable models, path-integral methods [9] and reference-based decompositions [10] are standard, and are available in consolidated form [11]. Broad surveys map the space [12]; Rudin [13] argues that for high-stakes decisions the post-hoc layer should be replaced by an intrinsically interpretable model, a position our Section V-C result speaks to directly and unfavourably.

Transfer of these methods to audio is contested. Haunschmid et al. [14] argue that audio explanations should satisfy a listenability criterion and that transplanting image-segmentation logic onto time–frequency representations yields components with no perceptual correlate; their remedy substitutes source-separated stems as the interpretable basis. Related formulations appear in [15] and [16], and Nasr et al. [17] extend the critique to speech emotion specifically, observing that a salient time–frequency region is not thereby an acoustically meaningful one. Source separation is inapplicable to monophonic single-speaker material, so Arm A adopts temporal windows and frequency bands — the representation under criticism — and mitigates rather than dismisses (Section IV-C). Arm B sidesteps the objection entirely by attributing over named acoustic descriptors in the tradition of GeMAPS [18] and openSMILE [19], where every unit of attribution has a phonetic reading by construction.

### C. Attribution stability as a validity requirement

The literature most directly load-bearing for this paper concerns the fragility of attribution itself. Ghorbani et al. [2] show that attributions can be substantially altered by perturbations below the threshold of perception while predictions and confidences remain stable. Adebayo et al. [20] demonstrate that certain widely deployed saliency methods are statistically independent of both model parameters and data-generating process, producing outputs with the surface form of explanation and none

of its content. Krishna et al. [21] document that different explanation methods routinely disagree on the same model, and that practitioners resolve the disagreement by unprincipled means.

The joint implication is that attribution stability must be measured against a null, not presumed. This requirement motivates the dither condition in Arm A and the decode-path control in Arm B, and it supplies the criterion against which we evaluate our own claims — including one we withdraw (Section VII-F).

#### D. Audio language models

Evaluation of large audio language models (LALMs) has expanded rapidly [22], [23], [24], with robustness work concentrating on adversarial manipulation and on naturally occurring corruptions such as noise and reverberation [25]. Two gaps persist. We identify no prior study that separates container from codec under a verified signal-invariance constraint. More elementary: across the corpus we surveyed, no study reports the container format in which audio was submitted. Corpus, prompt, and occasionally sampling rate are specified; the delivery representation is not. Lee et al. [23] document that cross-model comparison is already confounded by unreported variation in prompting and inference configuration; input representation belongs in the same category of under-reported experimental degrees of freedom.

On what these systems attend to, the evidence is mixed. Chen et al. [26] report systematic lexical dominance across six LALMs, with models defaulting to neutral when lexical content is affectively neutral and approaching chance in purely paralinguistic conditions. Zhang et al. [27] provide a counterweight at a scale inconsistent with chance-level competence, and identify zero-shot stochasticity in generative readouts as a first-order threat to evaluation validity. Our Arm A encounters both phenomena directly (Section IV-B).

#### E. Sensitivity, criterion, and calibration

Our principal Arm A finding is most precisely expressed in signal-detection terms. Sensitivity is the separability of the class-conditional response distributions, operationalised as the area under the ROC curve, and is invariant to threshold placement. Criterion is that placement, and can translate without any change in separability. The two are independently manipulable: Guo et al. [28] established that modern networks are frequently miscalibrated and that calibration error varies independently of accuracy, and related properties have been examined in language models [29]. A manipulation leaving accuracy invariant may still displace the operating point of any system that thresholds on posterior probability.

#### F. Where the gap lies

Coding effects are characterised for discriminative speech models but not for LALMs. Attribution fragility

is characterised for vision under adversarial perturbation but not for audio under ecologically routine degradation. No prior work isolates the container under a verified invariance, and none calibrates attribution divergence against a null perturbation. Critically for the present paper, no prior work places an opaque and a transparent model class on the identical manipulation, which is what licenses attributing a difference in explanation robustness to model family rather than to a difference in the signal chain.

### III. Study Design

#### A. Rationale

A single experiment cannot answer the question, because the two things we need to separate — what the codec does to the signal, and what the model does with the codec — are confounded within any one model class. An opaque model supports only perturbation-based attribution, whose estimator variance is unknown a priori; a transparent model supports exact attribution but constrains the hypothesis space to hand-designed features. Running both on the identical manipulation converts the difference between them from a nuisance into the measurement.

Fig. 1 shows the design. Two properties are load-bearing. First, single ancestry: every condition in both arms is generated from one canonical reference  $x$ , so the format contrast is the only systematic difference between conditions. Second, per-arm null calibration: each arm carries a manipulation that is informationally empty by construction, against which its effects are evaluated. Without the second property an observed divergence between two attribution maps is uninterpretable, because estimator variance and effect are not distinguishable.

#### B. Corpus and reference construction

Both arms draw on CREMA-D [30]: 7,442 clips from 91 actors, each delivering one of twelve fixed sentences in one of six acted emotions (anger, disgust, fear, happiness, neutral, sadness), with crowd ratings collected separately for voice-only, face-only, and audio-visual presentation.

The fixed lexical inventory is methodologically load-bearing. Holding the words constant across emotions and conditions removes the lexical channel as a source of discriminative variance, so any recovered signal must be carried by acoustic realisation. The same property makes the corpus maximally exposed to the lexical-dominance collapse Chen et al. [26] describe — a prediction Arm A confirms.

The canonical reference is 16 kHz monophonic 16-bit linear PCM, loudness-normalised to EBU R128 ( $I = -23$  LUFS,  $LRA = 7$ ,  $TP = -2$  dB). Coded conditions are produced from that reference by a single encoder invocation each, and all decoding in Arm B passes through one identical FFmpeg call so that the decoder never confounds the codec.

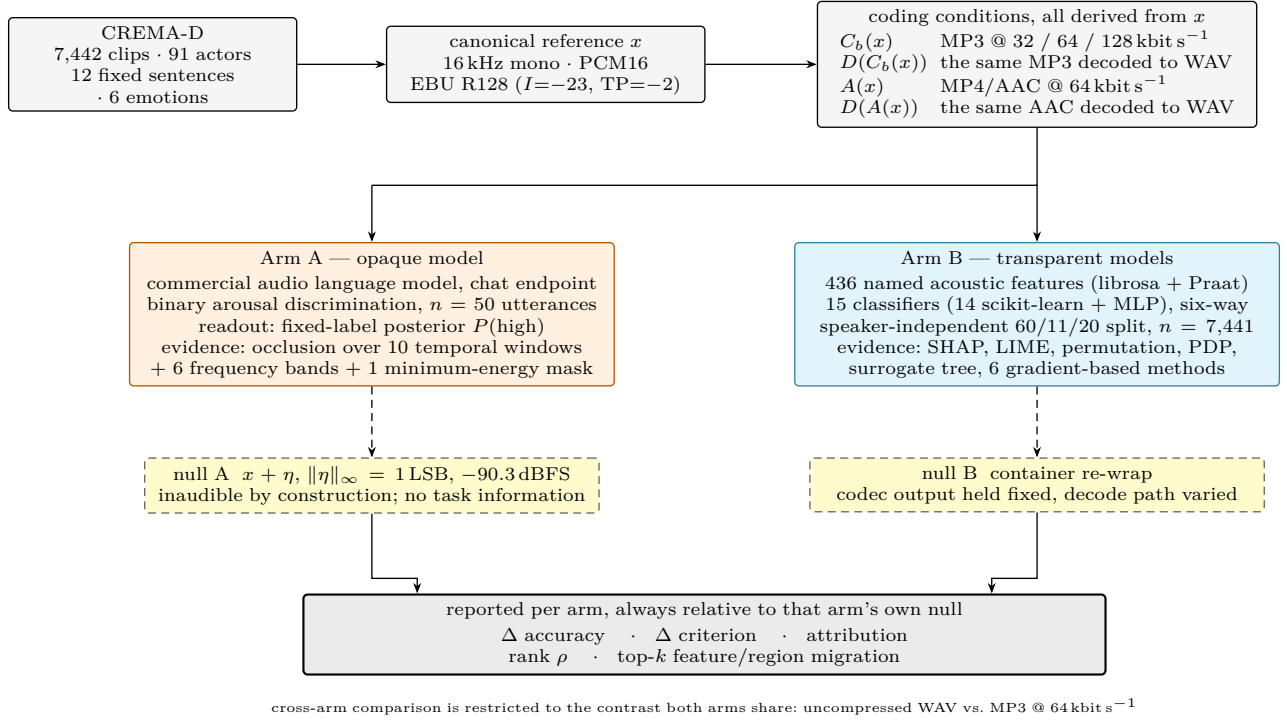


Fig. 1. The two-arm design. Every stimulus in both arms descends from a single canonical reference  $x$ , so no format contrast can be confounded by a divergent resampling or gain path. The two arms differ only in the model class placed downstream and in the attribution operator that class admits. Each arm carries its own null (dashed): an informationally empty waveform perturbation in Arm A, a container re-wrap that holds the codec output fixed in Arm B. All cross-arm statements are restricted to the contrast both arms share.

TABLE I

Contrasts and their estimands. The container contrast holds the waveform exactly constant, so any behavioural difference must arise downstream of decoding.

| Contrast             | What varies                   | Estimand            |
|----------------------|-------------------------------|---------------------|
| $D(C_b(x)) - x$      | waveform; container fixed     | codec effect        |
| $C_b(x) - D(C_b(x))$ | container; waveform invariant | container effect    |
| $C_b(x) - x$         | both                          | total format effect |
| $(x + \eta) - x$     | 1 LSB, inaudible              | null floor (A)      |

### C. The identification strategy

The container alters the byte layout; the codec alters the waveform. The design separates them by exploiting an equality that can be verified rather than assumed: the decoded PCM of a coded file and the PCM of that file re-wrapped as WAV are the same signal. Table I states the resulting contrasts.

In Arm A the invariance claim was verified directly: decoding both members of each pair through an identical fixed path and comparing SHA-256 digests of the resulting sample sequences yields 150/150 pairs bit-identical. In Arm B the corresponding control (AAC output routed through an MP4 container versus through a WAV container) is not bit-identical, because the WAV path passes through an additional int16 quantisation, but its median standardised feature difference is  $|\text{SMD}| = 7.6 \times 10^{-5}$  with only 12 of 436 features exceeding 0.05. That figure is the noise floor of Arm B: any effect larger than it is a codec

effect, not a decoding artefact.

### D. What the codec does to the signal

Fig. 2 characterises the manipulation in each arm's own instrumentation before any model is involved.

The two panels tell a consistent story from opposite directions. Perceptual bit allocation abandons the top octave, which Arm A sees as a 12.4 dB excess of band-limited distortion and Arm B sees as a single descriptor of that same octave moving by roughly 38 standard deviations. The distinction matters later: measured in training-set standard deviations, the shift is  $38.6\sigma$ , meaning the feature arrives at inference time in a range no model has ever seen.

AAC's contribution is different in kind. Its 32.68 ms of priming shifts frame alignment uniformly, which perturbs every first-order temporal derivative a little rather than destroying one band. The largest AAC shifts are consequently all delta-MFCC means (mfcc\_d1\_01\_mean,  $\text{SMD} = -4.16$ ). We flag a caveat here rather than in the limitations section, because it bears on how the number should be read: delta means sit near zero with small standard deviations, so their standardised shifts are inflated relative to their absolute change. The duration measurement is the harder evidence for the padding mechanism.

## IV. Arm A: An Opaque Audio Language Model

### A. System and constraints

The system under test is gpt-audio-1.5, accessed through an OpenAI-compatible Azure AI Foundry endpoint at

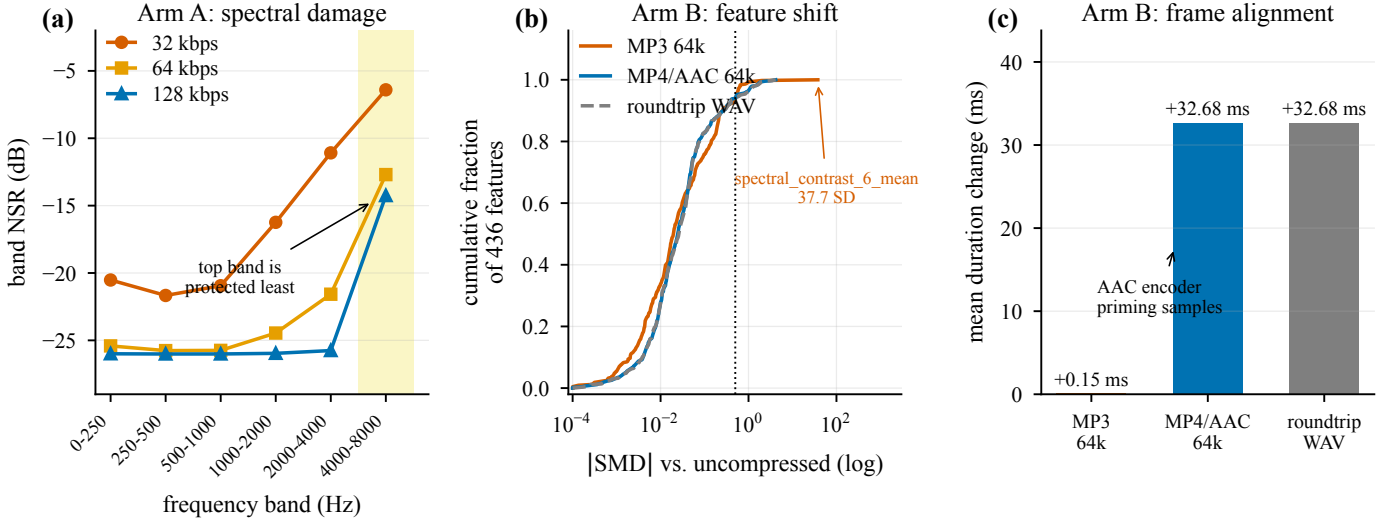


Fig. 2. The coding manipulation, measured independently in each arm. (a) Band-wise noise-to-signal ratio of the MP3 error signal (Arm A). Perceptual bit allocation protects the low-to-mid spectrum and quantises the top octave coarsely; at  $64 \text{ kbit s}^{-1}$  the 4 kHz to 8 kHz band carries 12.4 dB more distortion than the 500 Hz to 2000 Hz octave. (b) Empirical CDF of the absolute standardised mean difference of all 436 acoustic features against uncompressed audio, over 7,441 clips (Arm B). The median shift is negligible ( $\approx 0.02$ ); the tail is not. One feature, `spectral_contrast_6_mean`, moves by 37.7 pooled SD — it measures the 6 kHz to 8 kHz band that panel (a) shows the encoder abandoning. (c) AAC inserts encoder priming samples, lengthening every clip by 32.68 ms; MP3 adds 0.15 ms. The shift propagates into every frame-aligned dynamic feature.

temperature 0 with a fixed seed. Audio is transmitted as base64 bytes with a MIME format declaration; filenames and URIs are never transmitted, as either would leak condition identity into the prompt.

Two endpoint constraints shaped the design and must be reported because they bound the arm’s scope. The deployment rejects text-only requests, requiring an audio part on every call. It also rejects MP4/AAC payloads, accepting only WAV and MP3 — which fixes Arm A’s contrast as WAV versus MP3 and is the reason no AAC condition appears in this arm. Notably, submitting MP4 bytes under a wav format declaration elicits a different error from submitting an unsupported format string, indicating that the service validates container structure rather than trusting the declaration.

## B. Task selection and the readout

The initial formulation used the standard six-way categorical emotion task and failed: the model emitted *neutral* on 87.7% of trials and achieved 0.20 accuracy against a 0.167 chance rate, with 14/350 responses unparseable. This reproduces the lexical-dominance pattern Chen et al. [26] predict for affectively neutral lexical content, and it is fatal for our purposes: a task on which the system does not perform above chance cannot express a format effect, and attribution over a constant response is undefined.

We evaluated three binary reformulations on uncompressed audio, scored on bias-invariant discriminability rather than accuracy. Valence attained the highest nominal accuracy (0.81), which is precisely the pathology — it reproduces the majority-class base rate 34/42 exactly. Arousal partitions the six categories evenly (high: angry, fearful, happy; low: sad, neutral, disgusted), fixing chance

at 0.50, and corresponds to the affective dimension most directly encoded in intensity,  $F_0$ , and speech rate [31] — the correlates most exposed to coding. Arousal was selected.

The dependent variable is the renormalised posterior over the response alternatives extracted from returned log-probabilities. Critically, the statistic is  $P(\text{high})$  — the mass on a fixed class label, not on the correct one. The model emits *low* on 86.9% of trials; under a correct-label readout that bias alone inflates the statistic, whereas under a fixed-label readout a constant responder attains  $\text{AUC} = 0.5$  exactly, so bias is provably uninformative about the estimand. Extraction restricts to tokens with a non-null bytes field, because the returned distribution places most of its mass on fifteen distinct non-emittable special tokens that render identically as `<|end|>` and would otherwise deflate all posteriors.

Prompt selection required two rounds. The initial arousal prompt elicited 28% refusals of the form “please provide the audio you’d like me to listen to”. Placing the audio part before the text eliminated refusals but induced structured JSON emission with embedded chain-of-thought, destroying the single-token decision position the readout requires — and that reasoning quoted transcribed lexical content, which is independent evidence for lexical dominance. An explicit non-solicitation clause eliminated refusals without disturbing the response format. The final prompt yields 0% refusals, 0 unparseable responses, and 50/50 usable utterances.

TABLE II

Arm A discriminability by condition ( $n = 50$  utterances, fixed-label posterior, 10,000 bootstrap replicates).  $rt$  denotes an MP3 decoded back to WAV.

| Condition  | AUC   | 95 % CI        | $p$   | Acc. | mean $P(\text{high})$ |
|------------|-------|----------------|-------|------|-----------------------|
| ref        | 0.734 | [0.582, 0.864] | .0046 | 0.68 | 0.208                 |
| rt_mp3_32  | 0.717 | [0.562, 0.853] | .0088 | 0.64 | 0.158                 |
| rt_mp3_64  | 0.773 | [0.627, 0.896] | .0010 | 0.62 | 0.145                 |
| rt_mp3_128 | 0.710 | [0.554, 0.848] | .0110 | 0.64 | 0.153                 |
| mp3_32     | 0.726 | [0.578, 0.858] | .0062 | 0.60 | 0.159                 |
| mp3_64     | 0.746 | [0.587, 0.882] | .0030 | 0.58 | 0.144                 |
| mp3_128    | 0.758 | [0.608, 0.890] | .0018 | 0.62 | 0.147                 |

### C. Occlusion attribution

For utterance  $x$  and mask  $m_j$ , attribution for region  $j$  is the induced posterior decrement

$$a_j(x) = P(\text{high} | x) - P(\text{high} | m_j(x)), \quad (1)$$

so positive  $a_j$  indicates that suppressing region  $j$  removes evidence supporting the high response. The mask set comprises ten equal-duration temporal windows (each replaced by noise matched to the utterance’s long-term average spectrum, scaled to local RMS, with 10 ms raised-cosine crossfades), six band-stop filters on acoustically interpretable boundaries (0–250, 250–500, 500–1000, 1000–2000, 2000–4000, 4000 Hz to 8000 Hz), and one minimum-energy null mask that bounds the attribution attributable to the masking operation itself.

Masks are applied to the canonical reference and each format condition is regenerated from the masked signal, preserving single ancestry: the masked stimulus in condition  $c$  is the representation  $c$  would take had that region been absent, not a post-hoc corruption of an already-degraded signal.

In total the arm consumed 8,842 queries (8,016 unique calls, 0 errors); responses are content-addressed by a digest over deployment, audio bytes, prompt, and decoding parameters, rendering the arm re-executable from cache.

### D. Results

Fig. 3 and Table II give the three-way outcome.

1) Sensitivity is invariant: Every bootstrap interval covers the uncompressed estimate of 0.734. The accuracy decrement from 0.68 to 0.58 at  $64 \text{ kbit s}^{-1}$  does not attain significance: McNemar’s exact test [32] on discordant pairs yields 6 utterances correct only under ref against 1 correct only under mp3\_64 ( $p = .125$ ); at 32 and  $128 \text{ kbit s}^{-1}$ , 5 versus 1 ( $p = .219$ ) and 4 versus 1 ( $p = .375$ ). The sign is consistent across the ladder, which suggests a genuine small effect, but the design is underpowered at  $n = 50$  to resolve it. We report no detectable loss of sensitivity, not the absence of one.

2) The criterion is displaced: Marginal  $P(\text{high})$  declines from 0.208 to 0.144, a paired shift of  $-0.064$  ( $p = 6.8 \times 10^{-5}$ , Wilcoxon signed-rank). Decisively, both class-conditional distributions translate in the same direction and by comparable amounts — high-arousal utterances

TABLE III

Arm A effects relative to the 1 LSB null floor, at  $64 \text{ kbit s}^{-1}$ . Left: mean per-utterance  $|\Delta P(\text{high})|$ . Right: attribution rank concordance and top-3 region overlap, with  $p$  from a paired test against the null.

| Effect              | posterior       |                      | attribution |       |                      |
|---------------------|-----------------|----------------------|-------------|-------|----------------------|
|                     | mean $ \Delta $ | vs. floor            | $\rho$      | top-3 | $p$ vs. null         |
| null (1 LSB dither) | 0.0459          | $1.00\times$         | 0.699       | 0.567 | —                    |
| container           | 0.0379          | <b>0.83</b> $\times$ | 0.670       | 0.553 | .478                 |
| codec               | 0.0673          | $1.47\times$         | 0.491       | 0.413 | $4.2 \times 10^{-8}$ |
| total               | 0.0740          | $1.61\times$         | 0.492       | 0.453 | $2.7 \times 10^{-7}$ |

from 0.366 to 0.252, low-arousal from 0.050 to 0.036 — leaving their separation intact. In signal-detection terms this is a criterion shift with sensitivity held constant. It is a calibration phenomenon in the sense of Guo et al. [28], and it is by construction invisible to any accuracy-based metric.

3) The container contributes nothing: Table III normalises every effect by the measured null floor. Substituting an MP3 for its bit-identical decoded WAV displaces the posterior less than adding one least significant bit of noise. The correct inference is not that the container has a small effect but that it has no identified effect: the model exhibits mild generic sensitivity to waveform perturbation, and the container difference lies entirely within that envelope.

4) Attribution is substantially reordered: Three inferences follow from the right-hand block of Table III. The estimator floor is  $\rho = 0.699$ , not unity: an inaudible perturbation already dissipates roughly 30% of the rank ordering, so occlusion attribution on this system is substantially unstable and any format-effect claim advanced without this calibration would conflate estimator variance with effect. The codec effect nonetheless exceeds that floor decisively ( $\rho = 0.491$ , displacement 0.208,  $p = 4.2 \times 10^{-8}$ ), corroborated by an independent set-overlap statistic — fewer than half of the three most influential regions are preserved. The container effect does not ( $p = .478$ ), consistent with the posterior-domain result.

5) Why sensitivity survives: Fig. 4 supplies the mechanism. Model reliance concentrates unambiguously in the 500 Hz to 2000 Hz octave — the region carrying the bulk of speech energy and the first two formants, and the canonical locus of vocal effort. That a model-agnostic perturbation procedure recovers an acoustically principled region rather than an arbitrary one is itself evidence that the attribution operator is measuring something structured; fourteen of sixteen regions yield positive attribution, the theoretically expected direction.

The codec largely spares the discriminative evidence, which preserves sensitivity, while perturbing the periphery enough to reorder the attribution profile. A further observation supports treating signal fidelity and behaviour as separate quantities: band-wise distortion decreases monotonically with bitrate in every band, yet the codec effect on  $P(\text{high})$  is  $-0.050$ ,  $-0.063$ ,  $-0.055$  at 32, 64,  $128 \text{ kbit s}^{-1}$  with no monotone trend. The absence of dose-

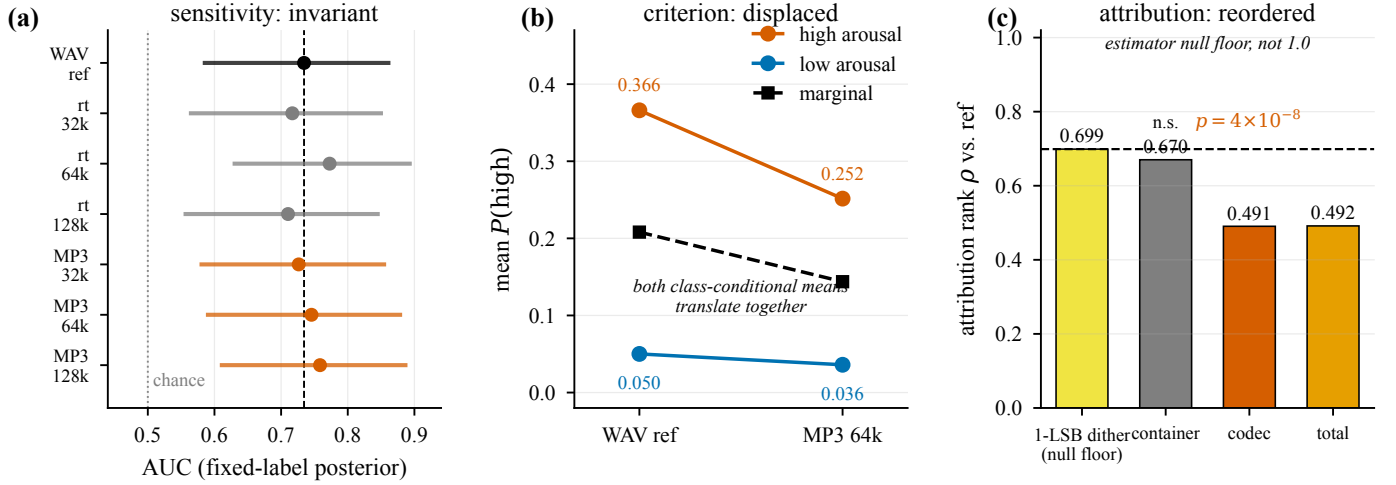


Fig. 3. Arm A: coding preserves sensitivity, displaces the criterion, and reorders attribution. (a) Discriminability with bootstrap intervals (10,000 replicates); every interval covers the uncompressed estimate. (b) Both class-conditional means translate downward by comparable amounts, leaving their separation intact — the signature of a criterion shift rather than a loss of sensitivity. (c) Attribution rank concordance against the uncompressed condition. The 1LSB dither establishes that this estimator dissipates  $\approx 30\%$  of the rank ordering under an informationally empty perturbation; the codec effect is evaluated against that floor, not against unity.

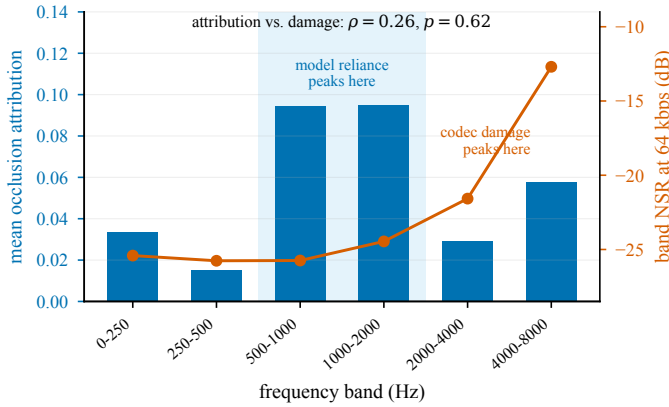


Fig. 4. Arm A: the spectral region the model relies on is among the least damaged. Bars, mean occlusion attribution on uncompressed audio; line, band noise-to-signal ratio of the coding error at  $64 \text{ kbit s}^{-1}$ . The rank correlation between reliance and damage is weak and non-significant, indicating no systematic alignment between what the codec destroys and what the model uses.

response is not a failed manipulation; it independently reproduces the central claim of Wu et al. [4].

## V. Arm B: Interpretable Models Over Named Features

### A. Feature representation

Each file is reduced to 436 named descriptors in four families, every frame-level contour collapsed to eight order statistics (mean, median, standard deviation, IQR, min, max, skew, kurtosis):

- spectral (114): centroid, bandwidth, rolloff (85/95 %), flatness, flux, entropy, slope, zero-crossing rate, RMS, seven-band spectral contrast, eight band-energy ratios, and explicit high-frequency survival ratios above 4 and 6 kHz;
- cepstral (240): 20 MFCCs with  $\Delta$  and  $\Delta\Delta$ ;

- prosodic (56), computed in Praat via Parselmouth [33]:  $F_0$  statistics and slope, jitter (local, RAP, PPQ5, DDP), shimmer (local, dB, APQ3, APQ5, APQ11, DDA), HNR, formants  $F_1$ – $F_5$  with bandwidths, intensity, voiced-segment timing, and smoothed cepstral peak prominence (CPPS);
- chroma (24) and two global descriptors (duration, peak amplitude).

Spectral and cepstral features are computed with librosa [34] at 512-point FFT, 160-sample hop, 400-sample window. Chroma tuning is pinned to zero rather than estimated, because an estimated tuning could itself shift under compression and would silently become part of the effect being measured.

The design choice worth defending is that these features are named. An attribution over prosody\_cppis is a claim about breathiness; an attribution over a learned embedding dimension is not a claim about anything. This is what makes the enter/leave analysis of Section V-E3 possible, and it is the analytical capability Arm A structurally lacks.

Extraction over  $7,442 \times 4 = 29,768$  files failed on exactly one clip in all four conditions, 1076\_MTI\_SAD\_XX, which the corpus documentation lists as having no audio. The remaining 7,441 clips form the analysis set.

### B. Protocol

Actors are partitioned 60/11/20 into train, validation, and test with sex balance preserved, yielding 4,905/896/1,640 clips. Speaker-independent splitting is not a nicety here. A variance decomposition over the feature table gives mean  $\eta^2 = 0.178$  for speaker against 0.098 for emotion, with 340 of 436 features speaker-dominant; a random row split would leak voice identity into the test set and inflate every score reported below.

Fourteen scikit-learn classifiers spanning tree ensembles [35], [36], kernel machines [37], linear and discriminant



TABLE IV

Arm B, trained and tested on uncompressed audio, 20 held-out speakers, six-way. Chance = 0.167; human voice-only = 0.455.

| Model                | Acc.  | Balanced acc. | Macro $F_1$ | $\kappa$ |
|----------------------|-------|---------------|-------------|----------|
| ANN (PyTorch)        | 0.599 | 0.599         | 0.596       | 0.518    |
| LightGBM             | 0.590 | 0.591         | 0.584       | 0.508    |
| SVM (RBF)            | 0.585 | 0.586         | 0.581       | 0.502    |
| LDA (shrinkage)      | 0.585 | 0.585         | 0.580       | 0.502    |
| Linear SVM           | 0.583 | 0.584         | 0.577       | 0.500    |
| XGBoost              | 0.581 | 0.582         | 0.575       | 0.497    |
| MLP (scikit-learn)   | 0.573 | 0.574         | 0.569       | 0.488    |
| HistGradientBoosting | 0.573 | 0.573         | 0.566       | 0.487    |
| Logistic regression  | 0.565 | 0.567         | 0.561       | 0.478    |
| Random forest        | 0.532 | 0.535         | 0.517       | 0.439    |
| AdaBoost             | 0.520 | 0.521         | 0.514       | 0.423    |
| Extra trees          | 0.516 | 0.519         | 0.496       | 0.420    |
| $k$ NN ( $k = 15$ )  | 0.477 | 0.479         | 0.466       | 0.373    |
| Gaussian NB          | 0.424 | 0.430         | 0.379       | 0.312    |
| Decision tree        | 0.390 | 0.391         | 0.382       | 0.268    |
| Dummy (majority)     | 0.171 | 0.167         | 0.049       | 0.000    |

models, instance-based and probabilistic baselines [38], plus a PyTorch MLP — fifteen trained models in all — (512-256-128 with batch normalisation and dropout 0.3, AdamW with cosine schedule, class-weighted cross-entropy with label smoothing, early stopping on validation UAR), are trained on the uncompressed condition and evaluated on all four. Item alignment is enforced: the same clips appear in every condition, so a cross-format comparison differs only in format.

For reference throughout: CREMA-D’s own crowd raters, hearing audio alone, matched the actor’s intended emotion on only 45.5% of clips, with per-class recall ranging from 0.966 (neutral) to 0.164 (sad). Models trained on the intended label can exceed this because they exploit systematic acting cues human listeners do not consciously decode; the figure is reported as context for absolute accuracy, not as a target.

### C. In-format performance

Table IV gives the uncompressed leaderboard. Two points matter for what follows, and neither is about the winner.

First, a single decision tree — the one model a human can read end to end — reaches 0.391 balanced accuracy against 0.599 for the network. Intrinsic interpretability costs a third of the achievable performance on this task. That is an argument against Rudin’s prescription [13] in this specific setting, and in favour of post-hoc attribution over a strong model; Section V-E2 then shows what that trade buys and what it costs.

Second, the spread between the top eight models is under three points, so the leaderboard ordering carries little information. The differences that matter in this paper are 10 to 38 points and appear only when the format changes.

### D. Cross-format performance

Table V and Fig. 5 report what happens when a model trained on clean audio is served compressed audio. The

result is categorical rather than graded. Every tree-based model sits within  $\pm 0.014$  of its clean score on MP3. Every model that consumes features as continuous magnitudes loses between 12 and 38 points.

The mechanism is legible because the features are named. A decision tree asks whether `spectral_contrast_6_mean` exceeds a threshold near 17; the answer is unchanged whether the feature reads 16.4 or 46.4. A linear or kernel model multiplies that  $38\sigma$  excursion straight into its decision function. This is textbook covariate shift [39], but with an unusual property: the shifted variable is not a proxy for anything the label depends on, so the shift is pure contamination.

MP4/AAC behaves differently, costing everyone something and nobody very much (0.7 to 11 points regardless of family), consistent with the diffuse padding mechanism of Fig. 2(c). Container routing has no material effect: across the fifteen models the median difference between roundtrip and `mp4_aac64` is 0.003 balanced accuracy and the maximum is 0.010 ( $k$ NN), against codec effects one to two orders of magnitude larger. For a feature pipeline that decodes both itself, this contrast is a null by construction — and confirming the null is exactly what licenses attributing everything else to the codec.

One further observation constrains the available remedies. Trained on `mp4_aac64` instead of `ref`, the RBF-SVM scores 0.572 on AAC and logistic regression 0.570, against 0.586 and 0.567 clean-on-clean. The emotional information survives  $64\text{ kbit s}^{-1}$  AAC essentially intact; what fails is the train/serve mismatch. But training on AAC does not rescue MP3 — the SVM falls to 0.167, exactly chance — so matched-format training is a format-specific fix, not a general immunisation.

### E. What the models attribute to

Attribution was computed on the held-out speakers using TreeSHAP for the ensembles [7], LinearSHAP for the linear models, LIME for local surrogates [5], permutation importance, partial dependence, a depth-4 global surrogate tree, and, for the network, six methods from Captum [11]: Saliency, Input $\times$ Gradient, Integrated Gradients [9], DeepLIFT [10], GradientSHAP, and Feature Ablation.

1) Convergent evidence across families: On uncompressed audio, five model families and four attribution algorithms select the same acoustic evidence. The top SHAP features for XGBoost and LightGBM are identical in the first five (`mfcc_d1_00_std`, `mfcc_d2_00_std`, `duration_s`, `prosody_intensity_std`, `mfcc_00_std`); the decision tree leads with `prosody_intensity_std`; and all six gradient methods on the network return `mfcc_d1_00_std`, `mfcc_d2_00_std`, `prosody_cpps`, and `duration_s`. The two leaders are the standard deviations of the first and second derivative of MFCC-0, which is to say how variable the rate of change of overall loudness is — a direct formalisation of vocal agitation. Per-emotion attributions are equally sensible: CPPS (a breathiness correlate) drives both sadness and



TABLE V

Arm B, all models trained on uncompressed audio and served compressed. Balanced accuracy on 20 held-out speakers. Models are grouped by how they consume a feature value. roundtrip is AAC output re-wrapped as WAV — the container control, whose difference from mp4\_aac64 has a median of 0.003 and a maximum of 0.010 (kNN) across the fifteen models.

|            | Model                | ref   | mp3_64 | $\Delta$ MP3 | mp4_aac64 | $\Delta$ AAC | roundtrip |
|------------|----------------------|-------|--------|--------------|-----------|--------------|-----------|
| thresholds | LightGBM             | 0.591 | 0.586  | −0.005       | 0.559     | −0.032       | 0.562     |
|            | XGBoost              | 0.582 | 0.586  | +0.004       | 0.560     | −0.022       | 0.560     |
|            | HistGradientBoosting | 0.573 | 0.559  | −0.014       | 0.530     | −0.043       | 0.533     |
|            | Random forest        | 0.535 | 0.532  | −0.003       | 0.522     | −0.013       | 0.522     |
|            | AdaBoost             | 0.521 | 0.511  | −0.010       | 0.487     | −0.034       | 0.484     |
|            | Extra trees          | 0.519 | 0.521  | +0.002       | 0.512     | −0.007       | 0.507     |
|            | Decision tree        | 0.391 | 0.382  | −0.009       | 0.367     | −0.023       | 0.371     |
| multiplies | kNN                  | 0.479 | 0.362  | −0.118       | 0.467     | −0.012       | 0.457     |
|            | Gaussian NB          | 0.430 | 0.277  | −0.152       | 0.437     | +0.008       | 0.433     |
|            | Linear SVM           | 0.584 | 0.405  | −0.179       | 0.471     | −0.113       | 0.462     |
|            | MLP (scikit-learn)   | 0.574 | 0.383  | −0.191       | 0.491     | −0.083       | 0.494     |
|            | ANN (PyTorch)        | 0.599 | 0.354  | −0.246       | 0.513     | −0.087       | 0.512     |
|            | LDA                  | 0.585 | 0.338  | −0.247       | 0.507     | −0.079       | 0.502     |
|            | Logistic regression  | 0.567 | 0.233  | −0.334       | 0.472     | −0.095       | 0.470     |
|            | SVM (RBF)            | 0.586 | 0.203  | −0.383       | 0.499     | −0.087       | 0.497     |
|            |                      |       |        |              |           |              |           |

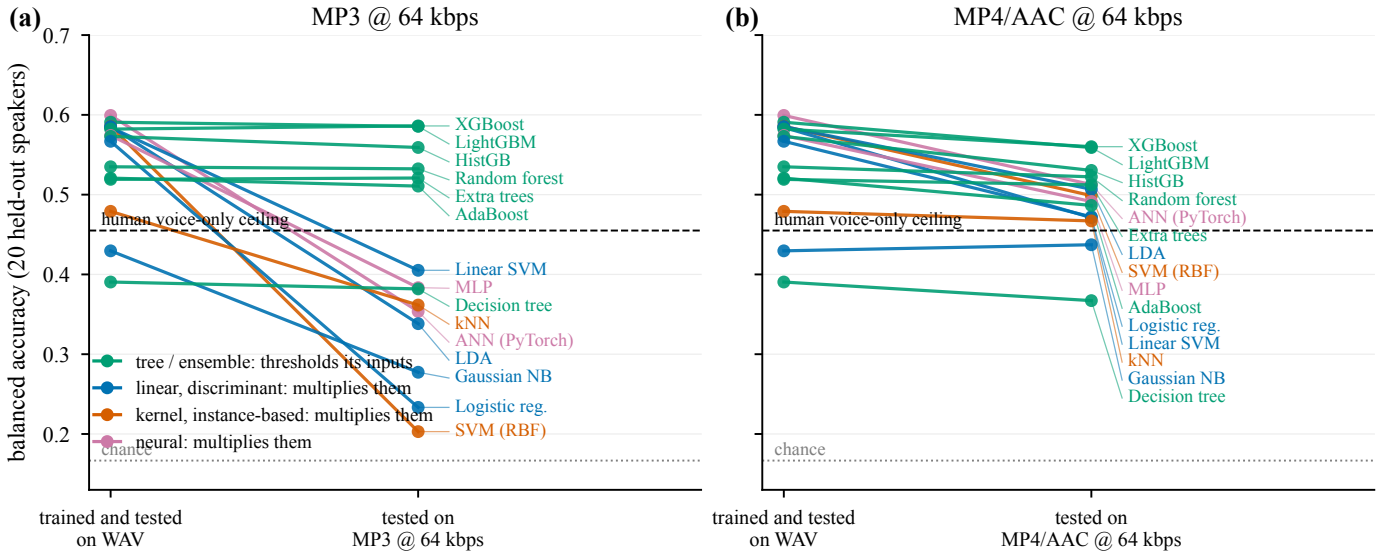


Fig. 5. Robustness to lossy coding is determined by how a model consumes its inputs, not by how accurate it is. (a) Under MP3 at  $64 \text{ kbits}^{-1}$  the tree family is flat while the magnitude-consuming family collapses through the human ceiling toward chance. (b) Under MP4/AAC the same models degrade broadly and mildly, consistent with a diffuse frame-alignment perturbation rather than a single-band blowout. All models were trained on uncompressed audio only.

neutrality,  $F_0$  median drives fear, and  $F_2$  movement drives disgust. The network was told none of this.

2) The methods disagree with each other: Fig. 6(a) is the uncomfortable result. SHAP agrees closely with a model’s intrinsic importance for the simpler models ( $\rho = 0.98$  for both the decision tree and logistic regression), but LIME and permutation importance are close to unrelated to it. On the decision tree, LIME and SHAP rank features at  $\rho = 0.008$  — no relationship at all — while still sharing 10 of their top 20. This reproduces the disagreement problem Krishna et al. [21] document, on a model simple enough that no approximation error can be blamed.

Panel (b) sharpens the point rather than softening it. The six gradient-family methods on the network agree almost perfectly ( $\rho \geq 0.94$  pairwise, with Inte-

grated Gradients and DeepLIFT at 0.998). High agreement within a method family is therefore not evidence of correctness — these methods share a mathematical lineage, and their consensus reflects that lineage. The practical consequence is that a single-method XAI claim about which acoustic features matter is not reproducible across methods. Only features surviving all four independent views should be reported as robust; here that set is mfcc\_d1\_00\_std, prosody\_intensity\_std, duration\_s, prosody\_cpps, and prosody\_f0\_median.

Two further calibrations belong with this result. A depth-4 global surrogate reproduces its target model’s predictions only 57 to 74 % of the time (random forest 73.9 %, LightGBM 63.4 %, XGBoost 62.1 %, SVM 58.7 %, logistic regression 57.1 %), so surrogate rules summarise a majority of a model, not a model; reporting them without

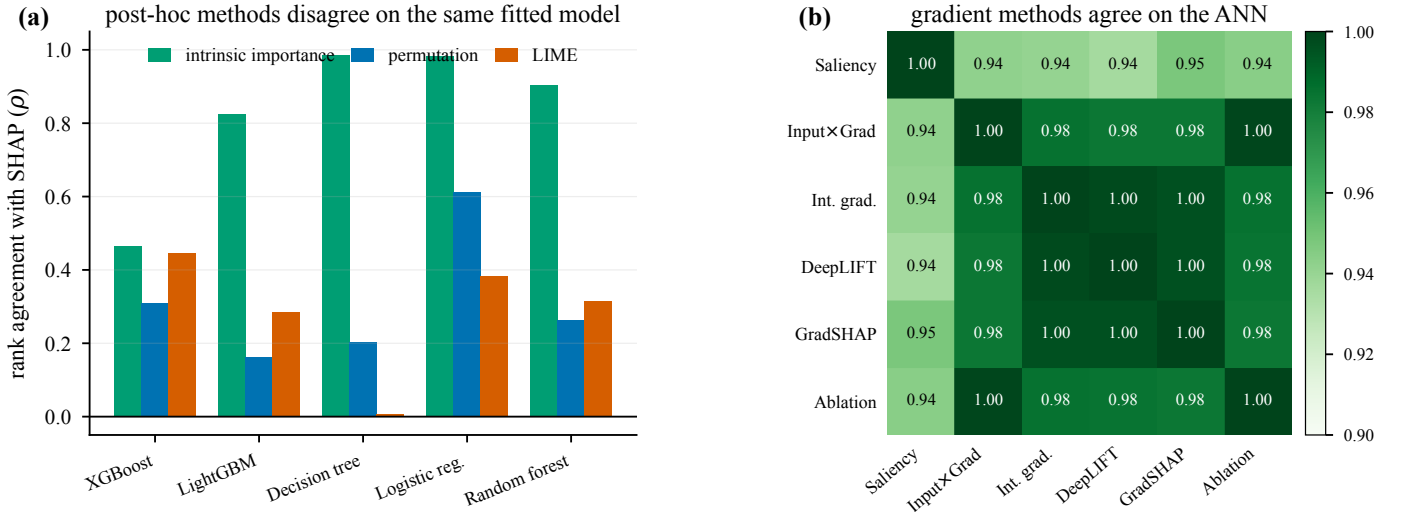


Fig. 6. Explanation-method agreement, measured as Spearman rank correlation over all 436 features. (a) Post-hoc methods applied to the same fitted tabular model. SHAP tracks intrinsic importance closely for the simpler models but LIME and permutation importance are nearly unrelated to it; on the decision tree, LIME and SHAP agree at  $\rho = 0.008$ . (b) The six gradient-family methods applied to the network agree almost perfectly ( $\rho \geq 0.94$  pairwise), which shows that the disagreement in (a) is a property of the method families rather than of this feature space.

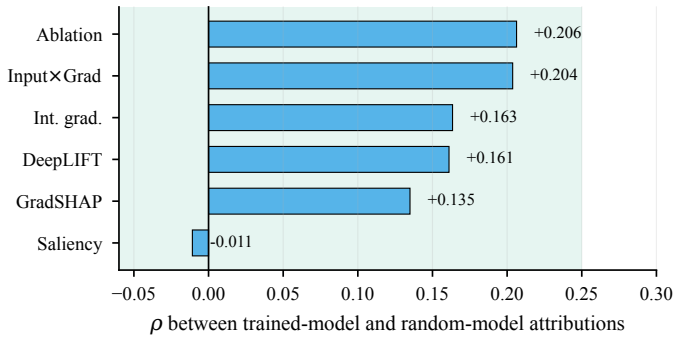


Fig. 7. Model-randomisation sanity check [20] for the six gradient-family methods. A method whose attributions correlate with those of a randomly initialised network of identical architecture is describing the input distribution rather than the model. All six sit near zero, so the failure mode does not apply here.

their fidelity would overstate what has been explained. And the network’s attributions pass the Adebayo model-randomisation check (Fig. 7): correlation between attributions from the trained network and a randomly initialised one of identical architecture ranges from  $-0.011$  to  $0.206$  across all six methods. These explanations describe the trained model, not merely the input distribution.

3) Attribution stability under coding: Table VI is the arm’s central explainability result. Under MP3, tree and linear attributions barely move; the network’s move substantially.

What replaces what is the whole story, and it is readable only because the features are named. Fig. 9(b) lists the migration. Under MP3 the features entering the network’s top 25 are almost all encoder artefacts — spectral\_flatness in five statistics, plus spectral\_contrast\_6\_mean — while the features leaving are genuine voice-quality and prosodic measures: prosody\_cpps, prosody\_f0\_skew, prosody\_f2\_mean,

TABLE VI  
Arm B attribution stability: Spearman  $\rho$  over all 436 features against the uncompressed condition, with the number of top-25 features retained in parentheses. The final column gives the accuracy change on MP3 for comparison. SHAP for all rows except the ANN, which uses Integrated Gradients.

| Model               | mp3_64     | mp4_aac64  | roundtrip  | $\Delta$ acc. MP3 |
|---------------------|------------|------------|------------|-------------------|
| Decision tree       | 0.995 (24) | 0.988 (21) | 0.988 (21) | −0.009            |
| XGBoost             | 0.991 (25) | 0.984 (22) | 0.984 (22) | +0.004            |
| Random forest       | 0.990 (25) | 0.976 (25) | 0.975 (25) | −0.003            |
| LightGBM            | 0.988 (23) | 0.983 (22) | 0.982 (22) | −0.005            |
| Logistic regression | 0.987 (22) | 0.942 (21) | 0.941 (21) | −0.334            |
| ANN (Int. grad.)    | 0.677 (11) | 0.891 (16) | 0.887 (16) | −0.246            |

prosody\_f2\_std, prosody\_f3\_bw\_median. The compressed model stops listening to the voice and starts listening to the encoder. Under AAC the same displacement occurs in milder form, with the padding-shifted mfcc\_d1\_\*\_mean entering and  $F_0$  statistics leaving.

4) A blind spot in global attribution metrics: The logistic regression row of Table VI is the most important negative result in this paper. Its SHAP ranking is essentially unchanged under MP3 ( $\rho = 0.987$ , 22 of 25 top features retained) while its balanced accuracy collapses from 0.567 to 0.233. The coefficients did not change; only the inputs went out of range, and a global mean-|SHAP| ranking is largely blind to that.

This is a caution about the method, not about the model. A global attribution-stability metric can report “nothing changed” while the classifier has effectively stopped working. Any deployment that monitors explanation stability as a proxy for model health must pair it with input-range monitoring or local, per-instance attribution; the global ranking alone will not raise the alarm.

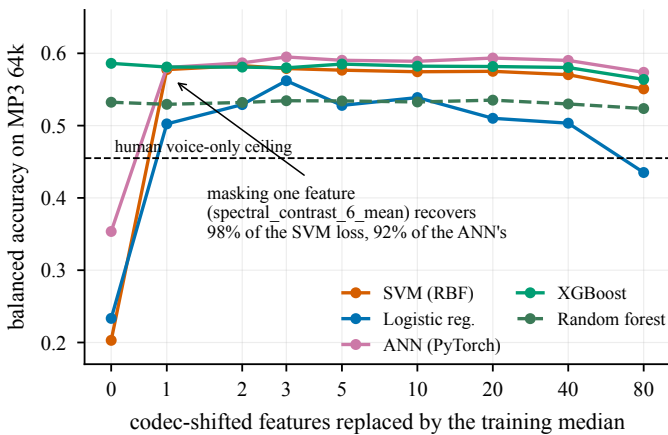


Fig. 8. Neutralisation on MP3 at  $64\text{ kbits}^{-1}$ : replacing the most codec-shifted features with their training medians. Masking one feature recovers almost the entire collapse for every affected model; the tree models were never affected and are flat throughout. Beyond roughly 20 features the masking begins to destroy real evidence. On uncompressed audio the same masking costs nothing out to 20 features, confirming these columns were never load-bearing evidence.

TABLE VII

Balanced accuracy on MP3 as the most codec-shifted features are replaced by training medians. Italic row: the model's own clean-audio score. Tree columns are flat because they had nothing to recover.

| Masked | SVM   | Log. reg. | ANN   | MLP   | XGBoost | Dec. tree |
|--------|-------|-----------|-------|-------|---------|-----------|
| clean  | 0.586 | 0.567     | 0.599 | 0.574 | 0.582   | 0.391     |
| 0      | 0.203 | 0.233     | 0.354 | 0.383 | 0.586   | 0.382     |
| 1      | 0.578 | 0.502     | 0.580 | 0.552 | 0.581   | 0.382     |
| 2      | 0.583 | 0.529     | 0.587 | 0.554 | 0.581   | 0.391     |
| 3      | 0.579 | 0.562     | 0.595 | 0.565 | 0.580   | 0.391     |
| 10     | 0.575 | 0.539     | 0.589 | 0.562 | 0.582   | 0.394     |
| 80     | 0.551 | 0.435     | 0.574 | 0.523 | 0.564   | 0.396     |

## F. Diagnosis and repair

Because Arm B's features are named, the failure is not merely observable but addressable. Features were ranked by how far each codec moves them in units of training standard deviation, then progressively replaced with training medians in both the compressed and the clean test set — the clean column controlling for whatever real signal the masking itself destroys.

Fig. 8 and Table VII give the outcome. The single masked feature is `spectral_contrast_6_mean`. Masking it alone recovers 98 % of the RBF-SVM's 38.3-point loss and 92 % of the network's 24.6-point loss; three features bring logistic regression to 0.562 against its 0.567 clean-audio score.

Two controls make this a diagnosis rather than a coincidence. On uncompressed audio the identical masking costs nothing out to 20 features — the network's clean control sits at 0.607–0.609, marginally above its 0.599 unmasked baseline — so these columns were never load-bearing evidence; they were the channel through which the encoder's fingerprint entered the model. And the AAC case behaves as its different mechanism predicts: no single feature dominates, with the network's best recovery

arriving at 10 features masked ( $0.513 \rightarrow 0.560$ ) and the SVM's at 40 ( $0.499 \rightarrow 0.542$ ).

The diagnosis is therefore covariate shift on a small set of identifiable descriptors, not loss of emotional information. That distinction is actionable, and the ordering of remedies follows from it: use a threshold-based model family, which is free; or clamp the top-band spectral descriptors, which recovers almost everything at the cost of one feature; or retrain on the target codec, which works but requires knowing the codec in advance and does not transfer between codecs.

## VI. Cross-Arm Synthesis

### A. Explanation displacement exceeds accuracy displacement

Fig. 9(a) places every model from both arms on one plane. Reading it requires one caution, which we state before the finding: the two arms' vertical coordinates are not the same measurement. Arm A's  $\rho$  is a per-utterance rank correlation over 16 spectro-temporal regions estimated by perturbation, with a measured null floor of 0.699; Arm B's is a global rank correlation over 436 features computed exactly, with a floor of unity. The figure places them on shared axes to compare magnitudes of displacement relative to each estimator's own floor, not to assert numerical commensurability.

With that caution, the pattern is consistent. Arm A occupies the top-left: displacement of 0.509 against a floor of 0.301, with an accuracy change that does not attain significance. This is the strong Ghorbani-style dissociation — the evidence moved and the decision did not.

Arm B does not reproduce the strong form. Every condition that moves its attributions also moves its accuracy, in the same direction and the same rank order across models. What does hold, and holds throughout, is the weaker and still consequential claim: the accuracy delta systematically understates the attribution delta. XGBoost loses 2.2 accuracy points on AAC and still swaps 3 of its top 25 features; the network loses 8.7 points and swaps 9. An accuracy-only benchmark would call the first negligible and the second modest.

Why the strong form appears in one arm and not the other is, we think, a property of the representation rather than of the model's sophistication. Arm B attributes over hand-designed descriptors, so a codec artefact enters as a specific out-of-range column that simultaneously corrupts the decision function of any model that multiplies it — coupling the two axes. Arm A's model learns its own representation end to end, and its attribution operator addresses spectro-temporal regions rather than features, so evidence can be reorganised across regions without the decision boundary being crossed. The dissociation is therefore more likely, not less, in exactly the deployed end-to-end systems where it is hardest to detect.

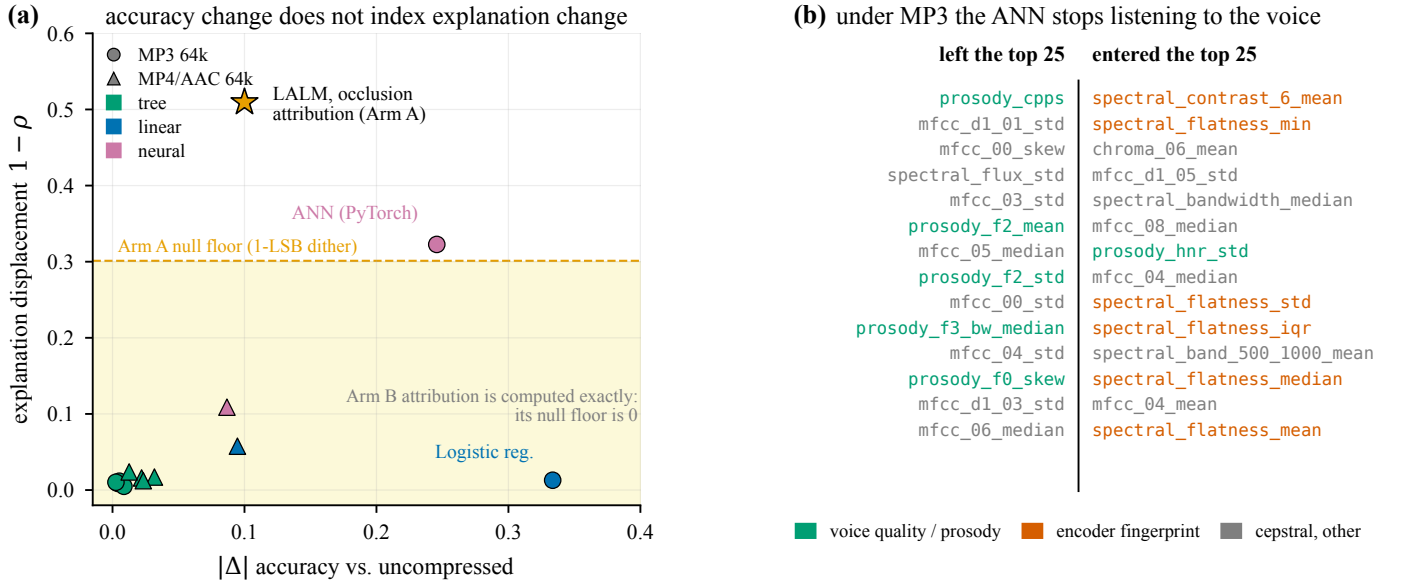


Fig. 9. (a) The accuracy-explanation plane for the shared WAV-vs-MP3 contrast, with MP4/AAC points from Arm B included as triangles. Points near the origin are robust on both axes; points high on the vertical axis have relocated their evidence. The two arms use different attribution estimators with different null floors — Arm A’s perturbation estimator dissipates 30% of the rank ordering under an informationally empty perturbation, whereas Arm B’s SHAP and Integrated-Gradients rankings are computed deterministically and have a floor of zero — so the shaded band applies only to the Arm A point. (b) The top-25 migration underlying the network’s displacement: voice-quality and formant descriptors leave, encoder-fingerprint descriptors enter.

TABLE VIII

Cross-arm summary on the shared contrast (uncompressed WAV versus MP3 at 64kbits<sup>-1</sup>). Attribution estimators differ between arms and are not numerically commensurable; each is reported against its own floor.

| Arm                                                 | Model class    | Attribution                                 | $\rho$      | Null floor   |
|-----------------------------------------------------|----------------|---------------------------------------------|-------------|--------------|
| B                                                   | Tree ensemble  | TreeSHAP (exact)                            | 0.988–0.995 | 1.000        |
| B                                                   | Linear         | LinearSHAP (exact)                          | 0.987       | 1.000        |
| B                                                   | Neural network | Integrated gradients                        | 0.677       | 1.000        |
| A                                                   | Audio LM       | Occlusion (estimated)                       | 0.491       | 0.699        |
| Container control (both arms): no identified effect |                |                                             |             |              |
| A                                                   | Audio LM       | Occlusion (estimated)                       | 0.670       | 0.699 (n.s.) |
| B                                                   | 15 models      | median $\Delta$ bal. acc. 0.003 vs. MP4/AAC |             |              |

B. Explanation robustness is ordered by model transparency

Table VIII arranges every model by attribution stability under the shared MP3 contrast. The ordering is striking and, to our knowledge, not previously reported: attribution stability decreases monotonically as the model becomes less inspectable. Tree ensembles  $\rho \geq 0.988$ ; linear models 0.987; the neural network 0.677; the audio language model 0.491 against a floor of 0.699.

We resist the temptation to read this as a law. Three confounds are present: attribution operator, feature representation, and model family all covary along the ordering, and the two arms differ in task, sample size, and label space. What the ordering does establish is that model family is a first-order determinant of explanation robustness — within Arm B, where operator and representation are held fixed, the network’s  $\rho = 0.677$  against the trees’  $\geq 0.988$  is a clean within-arm contrast on identical inputs and splits.

C. One physical cause, two behavioural outcomes

The arms agree precisely on the physics and diverge on the consequence, and this is the result neither arm could have produced alone.

Both localise the damage to the same place. Arm A measures the coding error directly and finds 4 kHz to 8 kHz distorted 12.4 dB more than the 500 Hz to 2000 Hz octave. Arm B measures the induced feature shift and finds that the single most-displaced descriptor of 436 is the one summarising 6 kHz to 8 kHz, at 38.6 training standard deviations. Two entirely different instruments, one location.

The behavioural consequence then splits on how each model consumes that band. In Arm A the band carries only moderate reliance (attribution 0.058 against 0.095 for the mid-spectrum), so sensitivity survives and only the attribution profile reorganises. In Arm B the band enters as an explicit descriptor; models that threshold it are unaffected, and models that multiply it lose up to 38 points. The codec did the same thing in both cases. What differed was the consumer.

This also resolves an apparent tension with the literature. Reddy and Vijayarajan [1] report modest, front-end-dependent codec effects on SER; we report catastrophic effects for half our models. Both are correct: the magnitude of a codec effect is not a property of the codec, and reporting it without specifying the model family is reporting an average over a bimodal distribution.

D. The container is inert, twice

The most robustly replicated result in this study is a null. Arm A: substituting an MP3 for its bit-identical de-

coded WAV displaces the posterior by  $0.83\times$  the inaudible floor and leaves attribution statistically indistinguishable from the null ( $p = .478$ ). Arm B: re-wrapping AAC output as WAV moves the fifteen models by a median of 0.003 balanced accuracy, with a maximum of 0.010 — against MP3 codec effects reaching 0.383.

The two nulls have different epistemic status and both are worth having. Arm B’s is close to structural — once FFmpeg decodes both routes, the contrast is nearly guaranteed — and its value is as a pipeline validation that licenses attributing everything else to the codec. Arm A’s is a genuine empirical result, because an API-fed model has an opaque decode path that could in principle differ by container, and the endpoint’s container-structure validation shows it inspects the wrapper. That it nonetheless behaves identically is informative.

The practical consequence is favourable: format conversion between containers, absent re-encoding, appears behaviourally inert. The operational risk localises entirely to the codec, which simplifies the mitigation problem considerably.

## VII. Discussion

### A. Attribution stability is a reportable robustness property

The results support a specific and inexpensive recommendation. Systems that surface per-instance rationales should report attribution stability across the input representations they will actually encounter, alongside accuracy, and should report it against a null. The two quantities are not redundant: in Arm A they dissociate completely, in Arm B the accuracy figure understates the attribution figure at every point, and in one case (Section V-E4) the attribution figure understates the accuracy collapse. Neither dominates the other, which is precisely why both are needed.

### B. Null calibration is not optional

The estimator floor we measure in Arm A — an inaudible perturbation dissipating 30 % of attribution rank ordering — implies that any claim of the form “manipulation  $M$  alters the explanation” is uninterpretable without a null perturbation matched in every respect except informational content. We propose this as standard practice for perturbation-based audio interpretability. The cost is one additional condition; the benefit is that the claim becomes falsifiable, as Section VII-F demonstrates at our own expense.

### C. Operational implications of criterion displacement

Deployed systems frequently act on posterior probability rather than on the argmax: escalation thresholds, triage routing, confidence-gated human review. A uniform displacement of 0.064 in posterior mass will alter the set of instances crossing any fixed threshold, purely as a function of ingestion format, while accuracy-based monitoring registers nothing. This yields a directly

testable operational prediction: organisations operating heterogeneous ingestion paths should observe systematic differences in confidence distributions across those paths absent any difference in measured accuracy.

### D. Auditing and the archive problem

Per-instance rationales are increasingly surfaced to operators and auditors. Our results indicate that such rationales are not invariant across a transcoding step that may leave the decision invariant. Where an audit is conducted on archived compressed material while inference runs on uncompressed input — or the converse — the decisions will agree while the explanations will not, with under half the top-ranked regions in common in Arm A and fewer than half the top-25 features in common for the network in Arm B. An explanation is not reproducible unless the delivery format is part of the record.

### E. What named features buy

Arm B’s mitigation (Section V-F) has no analogue in Arm A, and the asymmetry is structural rather than incidental. Identifying `spectral_contrast_6_mean` as the contaminated channel required attribution over units with acoustic meaning; neutralising it required reaching into the representation. Neither is possible through a chat-completions endpoint, where the only available lever is the input waveform.

This is a narrower claim than Rudin’s [13]. We are not arguing that interpretable models should replace black boxes — Table IV shows the single interpretable model in our zoo costing a third of achievable performance, and the best model overall is a neural network. We are arguing that a named-feature front end retains a diagnostic and repair capability that end-to-end pipelines lack, and that in codec-heterogeneous deployments this capability has measurable value: 98 % of a 38-point loss, recovered by masking one column.

### F. Corrections and threats to validity

An earlier iteration of the Arm A study reported a container effect: MP3 submission diverged from bit-identical decoded WAV on 47/47 utterances, at approximately half the magnitude of the codec effect. This was advanced as the principal finding. The supporting argument was that byte-identical requests return bit-identical responses — verified over five repetitions — so the measurement floor is exactly zero and any non-zero divergence constitutes signal.

The argument is invalid in a way we believe generalises. Zero variance under byte-identical inputs does not bound the response to inputs that differ. The relevant floor is the divergence induced by an informationally null modification. Once measured, that floor (0.0459) exceeds the container effect within the same arm (0.0379). One qualification on scope: the earlier estimate was computed on a different task and readout scale, so the magnitudes

are not directly commensurable and this is not a numerical refutation. What it establishes is that the earlier study used an inappropriate control and therefore never demonstrated that its effect exceeded any floor. We withdraw the claim as unsupported rather than disproven. The general principle: a determinism check is not a null control, and conflating them manufactures findings.

A second correction concerns Arm B. The RBF-SVM has no exact explainer, so it was given KernelSHAP with 100 sampled coalitions over 436 features. KernelSHAP fits a weighted least-squares regression with one unknown per feature; with 100 coalitions for 436 unknowns that system is rank-deficient, and the result diverged — maximum mean |SHAP| of  $2.1 \times 10^{12}$  against a median of  $5.3 \times 10^{-4}$ , with entire feature families assigned exactly zero. The SVM's KernelSHAP ranking is excluded from every conclusion in this paper as an estimation artefact, and the implementation now raises when the coalition budget does not exceed the feature count. The SVM contributes accuracy and robustness results but no attribution-stability row; its trustworthy views are permutation importance and LIME, whose top features reproduce the consensus every other model reaches.

### VIII. Limitations

- 1) Arm asymmetry. The endpoint rejects MP4/AAC payloads, so Arm A covers MP3 only and Arm B covers both codecs. All cross-arm claims are restricted to the WAV-versus-MP3 contrast both arms share; MP4/AAC results are Arm B evidence and are labelled as such throughout. Closing this gap requires a different deployment, not a different analysis.
- 2) Attribution estimators are not commensurable across arms. Arm A estimates by perturbation over 16 regions with a measured floor of 0.699; Arm B computes exactly over 436 features with a floor of unity. The ordering in Table VIII confounds operator, representation, and model family. The clean contrast is within Arm B, where all three are held fixed.
- 3) Single system, single corpus, single language. Arm A uses 50 utterances from one commercial deployment; both arms use acted English affect from one corpus. CREMA-D's human voice-only ceiling of 45.5% shows the labels are themselves only loosely recoverable from audio, and absolute accuracies do not transfer to spontaneous speech.
- 4) Arm A's sensitivity decrement is unresolved. The accuracy decline is directionally consistent across the bitrate ladder but non-significant at every bitrate. Only the criterion shift and the attribution reordering are supported.
- 5) Effect magnitudes in Arm A are modest. The codec exceeds the null floor by  $1.47\times$  on the posterior measure, not by an order of magnitude. The attribution effect is much stronger ( $p = 4.2 \times 10^{-8}$ ), but the two should not be conflated.

- 6) Arm B uses one split, not repeated cross-validation. All results come from a single deterministic 60/11/20 actor partition. Differences of a point or two on the leaderboard should not be over-read; the codec effects are 10 to 38 points and are not at risk from split variance.
- 7) Two bitrates in Arm B. Only  $64\text{ kbit s}^{-1}$  MP3 and AAC were tested. The MP3 result is severe partly because  $64\text{ kbit s}^{-1}$  is aggressive for that codec; higher bitrates would likely show a gradient. Arm A's ladder covers 32/64/128 but is compressed — the 64-to-128 increment yields only 0.46 dB of SNR, because the psychoacoustic model saturates well below  $128\text{ kbit s}^{-1}$  at 16 kHz monophonic input.
- 8) Arm B discards time. Every frame contour is collapsed to eight order statistics, so no Arm B attribution localises when in the clip the evidence sits. Arm A's temporal masks address this, but only for the opaque model.
- 9) Arm B's attribution comparisons are global. Spearman  $\rho$  and top- $k$  overlap over mean-|attribution| are coarse instruments, and Section V-E4 shows they can miss a total accuracy collapse. A local, per-instance stability analysis would be a stronger instrument and is not attempted here.
- 10) Arm A's interpretable basis is contested. Temporal windows and spectral bands are the representation criticised by [14], [17]; source separation, their preferred alternative, is inapplicable to monophonic single-speaker material. We mitigate — bands follow acoustically interpretable boundaries, masks are loudness-matched and edge-ramped, and a minimum-energy null mask bounds the attribution attributable to masking itself — but we do not dismiss the objection.
- 11) Occlusion indexes functional dependence, not mechanism. It licenses no claims about internal computation.
- 12) Both arms are exploratory. Task, prompt, readout, and null control in Arm A were selected after inspecting data from earlier arms. Confirmatory status requires preregistered replication.

### IX. Future Work

The highest-value extension is cross-system generalisation: replication across multiple audio language models, commercial and open-weight. Open-weight systems would permit gradient-based attribution and direct inspection of hidden representations, adjudicating whether the reordering we observe reflects genuine reorganisation of internal processing or is confined to the output surface — and would close the arm asymmetry by admitting AAC payloads.

Three further directions follow from specific results. The criterion-shift mechanism yields a falsifiable prediction: any degradation attenuating high-frequency transient content should displace the criterion toward low, irrespective of whether it removes affective information, which is



testable with degradations matched on broadband SNR but differing in spectral tilt. A null floor of  $\rho = 0.699$  is high, and since estimator noise can only attenuate a true difference, averaging over randomised mask partitions would likely increase the measured codec effect. Finally, if coding operates as an approximately additive criterion perturbation, a per-format affine recalibration of the posterior should eliminate the shift; this is inexpensive to test and immediately deployable.

Beyond arousal, speaker identity, vocal stress, and voice quality are carried by precisely the spectral detail perceptual coding discards, and may exhibit the sensitivity loss arousal does not.

## X. Conclusion

We asked whether perceptual audio coding changes what a speech emotion model relies on, as distinct from whether it changes what the model concludes. Across two model classes at opposite ends of the transparency spectrum, sharing one canonical reference signal and one coding manipulation, the answer is that it changes the evidence more reliably than it changes the decision.

An audio language model retains its discriminative sensitivity under coding while displacing its decision criterion by 0.064 and reordering its attribution profile to  $\rho = 0.491$  against a measured null floor of 0.699. Fifteen interpretable classifiers and networks show a graded version of the same phenomenon: accuracy loss consistently under-reports attribution change, and explanation stability is ordered by model family — tree ensembles above 0.988, the network at 0.677 — on identical inputs and splits. The container is inert in both arms. And the mechanism is the same in both: coding abandons the top octave, which Arm A sees as 12.4 dB of excess band-limited distortion and Arm B sees as one named descriptor arriving 38.6 training standard deviations out of range.

Two things follow that we would ask reviewers of affective computing systems to adopt. Report the delivery format; it is an experimental degree of freedom currently going unrecorded. And when reporting that an intervention changed a model’s explanation, report the divergence induced by an intervention carrying no information, because without it the claim cannot be evaluated — as we demonstrate by withdrawing one of our own.

## Data Availability Statement

CREMA-D is publicly available from its authors under the terms stated in [30]. All derived artefacts are released with this paper: the stimulus construction scripts and condition manifests, the timestamped 436-feature table over 29,768 rendered files with its JSON schema, the fitted model pipelines, and the JSON/CSV result files from which every table and figure in this paper is generated. Arm A’s API responses are content-addressed by a digest over deployment, audio bytes, prompt, and decoding parameters, so that arm re-executes from cache without

further inference calls. The figure-generation script reads only these committed artefacts.

## Ethics Declaration

This study involves no human subjects and no personally identifying data. It uses a published corpus of consenting professional actors performing scripted sentences, collected and released for research use by its original authors under their own ethical approval [30]. No new recordings were made. We note that speech emotion recognition is a dual-use technology; the findings here concern robustness and explanation validity and are, if anything, cautionary about deploying such systems on heterogeneous audio without format-aware validation.

## Author Contributions (CRediT)

Akshath R: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

## Conflict of Interest Statement

The author declares no competing financial or non-financial interests. Arm A was conducted against a commercial third-party deployment accessed under standard paid API terms; the vendor had no role in the design, execution, analysis, or reporting of this study and did not review the manuscript.

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## Declaration of Generative AI Use

Generative AI tools were used to assist with code implementation, figure generation, and manuscript drafting. All experimental design decisions, all analyses, and all interpretations are the author’s. Every numerical value reported was produced by the committed pipeline and traced to its source artefact; every citation was verified against a Crossref DOI record or the arXiv API. The author takes full responsibility for the content.

Separately and distinctly: a commercial audio language model is the object of study in Arm A, not a tool used to produce this manuscript.

## References

- [1] A. P. Reddy and V. Vijayarajan, “Audio compression with multi-algorithm fusion and its impact in speech emotion recognition,” *International Journal of Speech Technology*, vol. 23, no. 2, pp. 277–285, 2020.
- [2] A. Ghorbani, A. Abid, and J. Zou, “Interpretation of neural networks is fragile,” arXiv:1710.10547, 2019.
- [3] D. Alvarez-Melis and T. S. Jaakkola, “On the robustness of interpretability methods,” arXiv:1806.08049, 2018.



- [4] H. Wu, X. Chen, Y.-C. Lin, K. Chang, J. Du, K.-H. Lu, A. H. Liu, H.-L. Chung, Y.-K. Wu, D. Yang, S. Liu, Y.-C. Wu, X. Tan, J. Glass, S. Watanabe, and H.-y. Lee, "Codec-SUPERB @ SLT 2024: A lightweight benchmark for neural audio codec models," arXiv:2409.14085, 2024.
- [5] M. T. Ribeiro, S. Singh, and C. Guestrin, "“Why should I trust you?”: Explaining the predictions of any classifier," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 1135–1144.
- [6] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," arXiv:1705.07874, 2017.
- [7] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," *Nature Machine Intelligence*, vol. 2, no. 1, pp. 56–67, 2020.
- [8] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Computer Vision – ECCV 2014*, ser. Lecture Notes in Computer Science, vol. 8689, 2014, pp. 818–833.
- [9] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," arXiv:1703.01365, 2017.
- [10] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," arXiv:1704.02685, 2017.
- [11] N. Kokhlikyan, V. Miglani, M. Martin, E. Wang, B. Alsallakh, J. Reynolds, A. Melnikov, N. Kliushkina, C. Araya, S. Yan, and O. Reblitz-Richardson, "Captum: A unified and generic model interpretability library for PyTorch," arXiv:2009.07896, 2020.
- [12] R. Guidotti, A. Monreale, S. Ruggieri, F. Turini, F. Giannotti, and D. Pedreschi, "A survey of methods for explaining black box models," *ACM Computing Surveys*, vol. 51, no. 5, pp. 1–42, 2018.
- [13] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, 2019.
- [14] V. Haunschmid, E. Manilow, and G. Widmer, "audiolIME: Listenable explanations using source separation," arXiv:2008.00582, 2020.
- [15] S. Becker, J. Vielhaben, M. Ackermann, K.-R. Müller, S. Lapuschkin, and W. Samek, "AudioMNIST: Exploring explainable artificial intelligence for audio analysis on a simple benchmark," arXiv:1807.03418, 2018.
- [16] T. Sotiropoulos, V. Lyberatos, O. Menis Mastromichalakis, and G. Stamou, "MusicLIME: Explainable multimodal music understanding," arXiv:2409.10496, 2024.
- [17] S. Nasr, Z. Ren, and D. Johnson, "Beyond saliency: Enhancing explanation of speech emotion recognition with expert-referenced acoustic cues," arXiv:2511.11691, 2025.
- [18] F. Eyben, K. R. Scherer, B. W. Schuller, J. Sundberg, E. André, C. Busso, L. Y. Devillers, J. Epps, P. Laukka, S. S. Narayanan, and K. P. Truong, "The Geneva minimalist acoustic parameter set (GeMAPS) for voice research and affective computing," *IEEE Transactions on Affective Computing*, vol. 7, no. 2, pp. 190–202, 2016.
- [19] F. Eyben, M. Wöllmer, and B. Schuller, "openSMILE: The Munich versatile and fast open-source audio feature extractor," in Proc. 18th ACM Int. Conf. Multimedia, 2010, pp. 1459–1462.
- [20] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," arXiv:1810.03292, 2018.
- [21] S. Krishna, T. Han, A. Gu, S. Wu, S. Jabbari, and H. Lakkaraju, "The disagreement problem in explainable machine learning: A practitioner's perspective," arXiv:2202.01602, 2022.
- [22] Y. Chu, J. Xu, Q. Yang, H. Wei, X. Wei, Z. Guo, Y. Leng, Y. Lv, J. He, J. Lin, C. Zhou, and J. Zhou, "Qwen2-Audio technical report," arXiv:2407.10759, 2024.
- [23] T. Lee, H. Tu, C. H. Wong, Z. Wang, S. Yang, and Y. Mai, "AHELM: A holistic evaluation of audio-language models," arXiv:2508.21376, 2025.
- [24] K. Luo, Z. Zhou, L. Wang, L. Lin, T. Shao, and Y. Zhang, "A survey of large audio language models: Generalization, trustworthiness, and outlook," arXiv:2605.20266, 2026.
- [25] K. Li, C. Shen, Y. Liu, J. Han, K. Zheng, X. Zou, L. Z. Wang, S. Zhang, X. Du, H. Luo, Y. Jin, X. Xing, Z. Ma, Y. Liu, Y. Zhang, J. Fang, K. Wang, Y. Yan, G. Deng, and X. Li, "AudioTrust: Benchmarking the multifaceted trustworthiness of audio large language models," arXiv:2505.16211, 2025.
- [26] J. Chen, Z. Guo, J. Chun, P. Wang, A. Perrault, and M. Elsner, "Do audio LLMs really LISTEN, or just transcribe? measuring lexical vs. acoustic emotion cues reliance," arXiv:2510.10444, 2025.
- [27] H. Zhang, H.-C. Chou, S. Narayanan, and T. Hain, "VoxEmo: Benchmarking speech emotion recognition with speech LLMs," arXiv:2603.08936, 2026.
- [28] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," arXiv:1706.04599, 2017.
- [29] S. Kadavath, T. Conerly, A. Askell, T. Henighan, D. Drain, E. Perez, N. Schiefer, and Z. Hatfield-Dodds, "Language models (mostly) know what they know," arXiv:2207.05221, 2022.
- [30] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "CREMA-D: Crowd-sourced emotional multi-modal actors dataset," *IEEE Transactions on Affective Computing*, vol. 5, no. 4, pp. 377–390, 2014.
- [31] J. A. Russell, "A circumplex model of affect," *Journal of Personality and Social Psychology*, vol. 39, no. 6, pp. 1161–1178, 1980.
- [32] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.
- [33] Y. Jadoul, B. Thompson, and B. de Boer, "Introducing Parselmouth: A Python interface to Praat," *Journal of Phonetics*, vol. 71, pp. 1–15, 2018.
- [34] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in Python," in Proc. 14th Python in Science Conference, 2015, pp. 18–24.
- [35] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [36] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 785–794.
- [37] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [38] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," arXiv:1201.0490, 2011, *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830.
- [39] J. G. Moreno-Torres, T. Raeder, R. Alaiz-Rodríguez, N. V. Chawla, and F. Herrera, "A unifying view on dataset shift in classification," *Pattern Recognition*, vol. 45, no. 1, pp. 521–530, 2012.